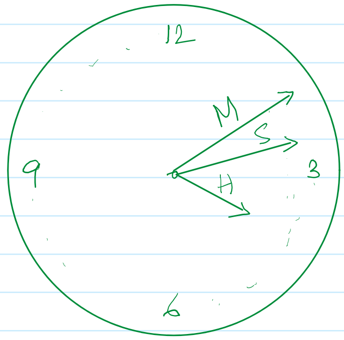
Nested loops

04 August 2026 18:10



60 sec = 1 min
60 min = 1 hour

for each hour
the minute needle move 60 steps
for each min
the seconds needle move 60 steps

The nested loop analogy is defined or understand by the clock.

(i) while loop (ii) for loop (iii) do... while loop

```
while (condition){
    while (condition){
        ....
    }
}
```

nested or inner loop

The loop will complete its execution for each step of outer loop.

```
for (        ){
    for (     ){
        ....
    }
}
```

```
while (      ){
    for (        ){
        ..
    }
}
```

```
for (          ){
    while (     ){
    }
}
```

```
do{
    for(        ){
    }
}while(     );
```

The nesting of loop can be any level.

```
for (       )
    for(       )
        for (    )
            while ( ){


            }
        }
    }
}
```

```
package Loops.NestedLoops;
import java.util.*;


/**
 * Write a description of class MultiplicationTable here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class MultiplicationTable
{
    public static void main(String args[]){
        int N, M, i, j, p;
        Scanner sc = new Scanner(System.in);
        System.out.print("\f======================================================\n");
        System.out.print("\nMultiplication Tables upto 10 terms for each numbers\n");
        System.out.print("\n======================================================\n");
        System.out.print("Enter starting value: ");
        N = sc.nextInt();
        System.out.print("Enter the end value: ");
        M = sc.nextInt();
        System.out.print("\n======================================================\n");
        for(; N <= M; N++){
            System.out.print("\n======Multiplication table of " + N+" ======\n");
            for(j = 1; j <= 10; j++){
                p = N * j;
                System.out.print("\n"+N+" * " + j + " = " + p);
            }
            System.out.print("\n");
        }
    }
}
```

Output:

```
======================================================

Multiplication Tables upto 10 terms for each numbers

======================================================
Enter starting value: 5
Enter the end value: 10

======================================================

======Multiplication table of 5 ======

5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
```

```
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50

======Multiplication table of 6 ======

6 * 1 = 6
6 * 2 = 12
6 * 3 = 18
6 * 4 = 24
6 * 5 = 30
6 * 6 = 36
6 * 7 = 42
6 * 8 = 48
6 * 9 = 54
6 * 10 = 60

======Multiplication table of 7 ======

7 * 1 = 7
7 * 2 = 14
7 * 3 = 21
7 * 4 = 28
7 * 5 = 35
7 * 6 = 42
7 * 7 = 49
7 * 8 = 56
7 * 9 = 63
7 * 10 = 70

======Multiplication table of 8 ======

8 * 1 = 8
8 * 2 = 16
8 * 3 = 24
8 * 4 = 32
8 * 5 = 40
8 * 6 = 48
8 * 7 = 56
8 * 8 = 64
8 * 9 = 72
8 * 10 = 80

======Multiplication table of 9 ======

9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
```

```
======Multiplication table of 10 ======

10 * 1 = 10
10 * 2 = 20
10 * 3 = 30
10 * 4 = 40
10 * 5 = 50
10 * 6 = 60
10 * 7 = 70
10 * 8 = 80
10 * 9 = 90
10 * 10 = 100
```

WAP to input number of rows and print the following patterns:
Input: Enter number of rows: 5

```
1.            1
            1 2
          1 2 3
        1 2 3 4
      1 2 3 4 5


2.            1
            2 1
          3 2 1
        4 3 2 1
      5 4 3 2 1
```